



Ranjan Jyoti Sarma

Senior Research Scientist-
Bioinformatics

PROFILE SUMMARY

1. Deep understanding of NGS Library preparation methods (Exome, RNASeq and NIPS).
2. Validated Illumina Onboard DRAGEN v4.13 for Clinical reporting of Whole Exome/Clinical Exome (SNV, SV and CNV) and BRCA Panels.
3. Pipeline development, streamlining in workflow management software like NextFlow.
4. Implementing pipeline in GNU Parallel, Log generation, resource optimization.
5. Variant database development using SQLite and implementing through OpenCRAVAT Annotation GUI variant annotation tool locally in server.
6. Developed HPO filtering plugin for OpenCRAVAT.
7. RShiny app to effectively handle variant annotated data.
8. Well versed with GATK, DRAGEN, Illumina BaseSpace and other popular NGS processing tools.

PERSONAL INFORMATION

- ✉ **Email**
cbiorjs@gmail.com
- ☎ **Mobile**
(+91) 97744-40639
- 📁 **Total work experience**
7 Years
(Academic: 6 Years, Industry: 1 Year)
- 🌐 **Social Link**
<https://in.linkedin.com/in/ranjanbinf>
- 🌐 **GitHub**
<https://github.com/biocoderdev>

KEY SKILLS

BASH, R, PYTHON, SQL, HTML, PHP, Flask

Computational Biology

Bioinformatics

NGS Data Analysis

RNA-Seq Data Analysis

Whole Exome/Genome Data Analysis

Clinical Data Analysis

OTHER SKILLS

Machine Learning Packages

- *Pandas
- *NumPy
- *Scikit-learn

OTHER PERSONAL DETAILS

Current City: New Delhi, India

EDUCATION

- 2024 M.Phil and Ph.D in Biotechnology
Specialization: Genomics
Mizoram University, Aizawl
- 2017 MS/M.Sc Bioinformatics
Pondicherry University
- 2015 B.Sc Biotechnology
North Eastern Hill University (NEHU)
- 2011 XII
Science AHSEC,
Assam
- 2009 Xth
SEBA, Assam

WORK EXPERIENCE

May 2024 -
Present

Senior Research Scientist-Bioinformatics
Dr Lal PathLabs

Experienced in handling large scale genomic data of different libraries. Perform in depth QC, perform automation in data analysis. Along with strong NGS library preparation concept, I carry robust experience in Illumina BaseSpace, GATK, DeepVariant, ANNOVAR, VEP, OpenCRAVAT, SNPeff including other popular genome analysis tools and databases. Strong attitude towards building annotation database, tools to provide better clinical insight and commercialization.

LANGUAGES

- English
- Hindi
- Assamese

Jun 2017 - Apr 2024

Project Research Fellow
Mizoram University
Junior Research Fellow, Project Associate- Data Analyst
(DBT-GenomeIndia)

Projects

Published as
Pre-print

**Automating Whole Exome Data Analysis for Variant
Discovery**
<https://github.com/biocoderdev/weap>

COURSES & CERTIFICATIONS

- Training in Linux

THESIS

- Ph.D**
Genetic Variants in Pharmacogenes in North Indian population and their association with Adverse Drug Reaction in Gastric Cancer
- M.Phil**
Detection of fusion genes and their expression in gastric cancer patients in Mizo population

PUBLICATION (9)

- Journal**
8 Articles
- Pre-print**
1 Article
- Poster (2) and Oral (1)**
2 International and 1National
- Google Scholar**
https://bit.ly/Ranjan_GS